

Programme	B. Sc. Mathematics Honours			
Course Code	MAT3MN205			
Course Title	OPTIMIZATION TECHNIQUES			
Type of Course	Minor			
Semester	III			
Academic Level	200 - 299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic understanding of linear algebra and introductory optimization concepts.			
Course Summary	This course provides a comprehensive exploration of linear programming and optimization techniques, focusing on graphical methods, the simplex method, and specialized problems like transportation and assignment. Students will gain practical skills in formulating, solving, and analyzing linear programming models, with applications in various optimization scenarios.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the fundamental properties and types of linear programming models, distinguishing between maximization and minimization models, and explain various methods used for solving linear programming problems including graphical methods.	U	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Apply the simplex method to solve both maximization and minimization linear programming problems, compare the graphical method with the simplex method in terms of efficiency and applicability, and demonstrate problem-solving skills through worked-out examples.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO3	Evaluate and solve transportation and assignment problems using specific techniques such as the North-West corner method, Least Cost cell method, Vogel's approximation method, and the Hungarian method, while also comparing the transportation model with general linear programming models.	An	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam

Detailed Syllabus:

Text book		Operations Research (2/e), P Rama Murthy, New Age International Publishers		
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I	Linear Programming Models: (Graphical Method)		10	17
	1	Section 2.1- Introduction, 2.2- Properties of Linear Programming Model		
	2	Section 2.3- Maximization Models		
	3	Section 2.4- Minimization Models		
	4	Section 2.5- Methods for the Solution of a Linear Programming Problem (up to Problem 2.9)		
	5	Section 2.5- Methods for the Solution of a Linear Programming Problem (From Problem 2.9)		
II	Linear Programming Models: (Simplex Method)		13	18
	6	Section 3.1- Introduction, 3.2- Comparison Between Graphical and Simplex Methods		
	7	Section 3.3- Maximisation Case		
	8	Section 3.4- Minimisation Case		
	9	Section 3.7- Minimisation Problems up to Problem 3.11		
	10	Section 3.7- Rest of the Section.		
III	Linear Programming Models: (Two Phase Simplex Method and Transportation Problem)		11	18
	11	Section 3.10- Artificial Variable Method or Two Phase Method up to Problem 3.19		
	12	Section 3.10- Rest of the Section.		
	13	Section 3.11- Degeneracy in Linear Programming Problems		
	14	Section 4.1 Introduction , 4.2 Transportation model		
	15	Section 4.3 – Comparison between Transportation model and general linear programming model		

	16	4.4- Approach to solution to a transportation problem by Transportation Algorithm up to and including 4.4.2	
IV		Linear Programming Models: (Transportation Problem and Assignment Problem)	14
		 Section 4.4.3- Basic feasible solution by North -West corner method	17
	18	Section 4.4.4- Solution by Least Cost cell method	
	19	Section 4.4.5- Solution by Vogel's approximation method	
	20	Section 4.4.6- Optimality test- Stepping stone method of optimality test only	
	21	Section 5.1 Introduction, 5.2 – The Problem	
	22	Section 5.4- Approach to solution-Hungarian method(Other methods of solution are optional)	
V		Open Ended Module	12
		Simplex method special Cases- Alternate solution. Unbound Solutions ,Problem with Unrestricted Variables	
		Transportation model- Modified distribution method	
		Game theory	

References :

1. KV Mittal and C Mohan, Optimization methods in Operations research and system analysis(3/e)
2. Kanti Swarup, PK Gupta and Manmohan, Operations Research(20/e)

Note: 1) Optional topics are exempted for end semester examination. (2) Proofs of all the results are exempted for external exam. (3) The 70 external marks are distributed across the first four modules as per the allocation specified in the syllabus.

Mapping of COs with PSOs and POs :

	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	3	2	3	2	3	1	2
CO 2	3	2	3	2	3	2	3	1	2
CO 3	3	2	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓